

Laser-Powered Vibrotactile Rendering

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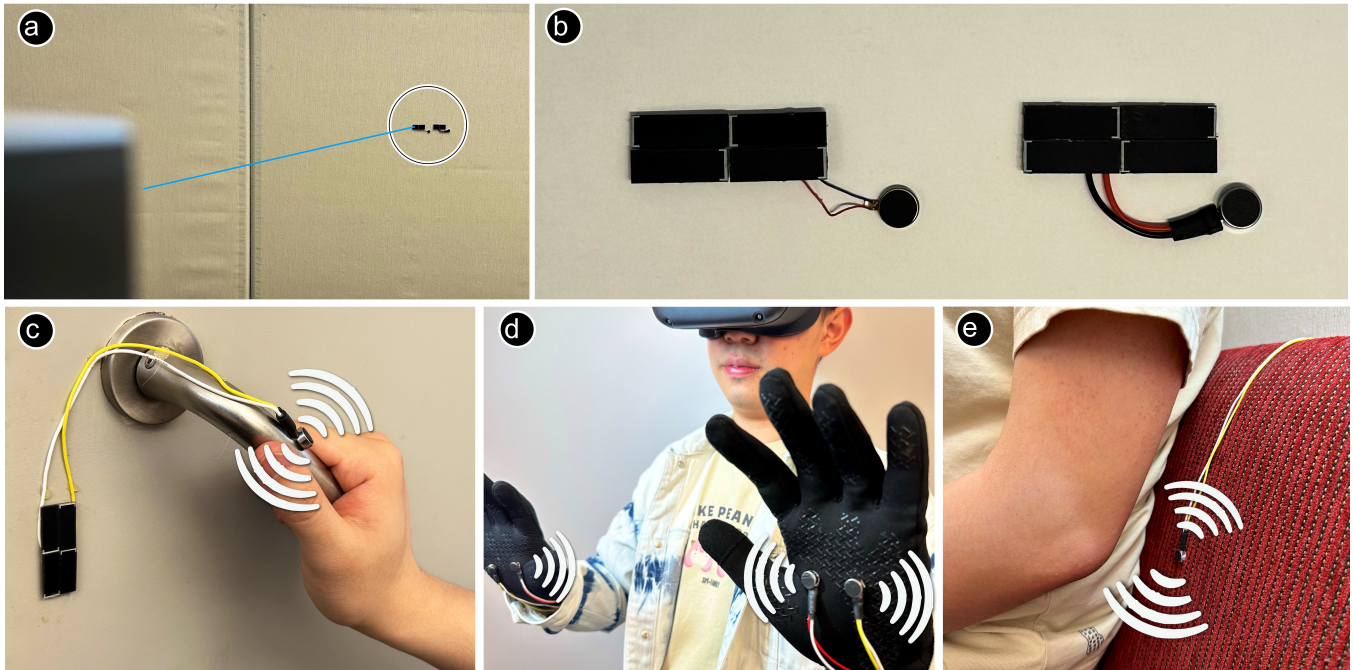


Figure 1: Top row: illustration of our battery-free, electronic-free vibrotactile device. (a) The device is powered and controlled by a laser, avoiding the need for frequent charging or battery replacement. (b) The device is made up of solar cells connected to a linear resonant actuator (LRA) or eccentric rotating mass (ERM) motor. Bottom row: application scenarios for prolonged device operation. (c) A door handle with ambient haptic feedback to show if the room is occupied. (d) Haptic gloves for immersive VR experiences (e) An ambient haptic display on a coach to remind a user of important upcoming meetings.

ABSTRACT

We investigate the feasibility of a vibrotactile device that is both battery-free and electronic-free. Our approach leverages lasers as a wireless power transfer and haptic control mechanism, which can drive small actuators commonly used in AR/VR and mobile applications with DC or AC signals. To validate the feasibility of our

method, we developed a proof-of-concept prototype that includes low-cost eccentric rotating mass (ERM) motors and linear resonant actuators (LRAs) connected to photovoltaic (PV) cells. This prototype enabled us to capture laser energy from any distance across a room. Through different vibration patterns rendered using either a single motor or two motors, we demonstrate the effectiveness of our approach in generating vibration patterns.

CCS CONCEPTS

• Human-centered computing → Haptic devices.

KEYWORDS

Haptics, Energy Harvesting, Virtual Reality

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1 INTRODUCTION

Small and lightweight electromagnetic actuators, such as LRAs and ERM motors, have widespread usage in AR/VR and ubiquitous computing applications for providing rich vibrotactile feedback. However, the requirement for regular battery charges or replacements presents a considerable challenge for their deployment in daily use cases that require extended system operation. In recent years, battery-free technologies have emerged as a promising solution, with actuators that operate using alternative sources such as light [1] or kinetic energy [3] exhibiting the potential to enable prolonged device operation. However, the incorporation of energy harvesting and power management components, in addition to the already cumbersome electronics and hardware, results in a trade-off that yields a bulky device form factor that may be intrusive when deployed on everyday objects or uncomfortable to wear on the human body.

In this paper, we examine the viability of developing a vibrotactile device comprising small ERM motors or LRAs without requiring any batteries or electronic components, such as microcontrollers, driver chips, or Wi-Fi modules. Our approach involves using a laser as a wireless power transfer and haptic control mechanism, enabling the actuators to be driven using DC (e.g., ERM) or AC (e.g., LRA) signals. The proposed method replaces the battery with small PV cells that harness laser energy at any distance across a room. Additionally, electronic components required for power management and motor control, including drivers, microcontrollers, and other hardware, are removed. Instead, the intensity or duty cycle of the laser is used to control the rendering of different vibrational patterns. The outcome is a simple and low-maintenance device architecture, as illustrated in (Figure 1a-b).

In contrast to alternative wireless charging approaches such as magnetic induction, lasers are capable of operating over long distances with minimal attenuation. Compared to far-field technologies including microwaves [5] and radio-frequency identification (RFID) [2], as well as optical methods utilizing visible light [4], lasers have high power density, leading to a smaller form factor for the haptic device that is crucial in daily applications. Additionally, lasers offer directional transmission, which enables the precise control and focused delivery of power to a particular device.

2 WORKING PRINCIPLE AND IMPLEMENTATION

Our system is a solar-powered actuator that uses a laser source to control vibration motors in AR/VR and ubiquitous computing applications, eliminating the need for batteries, power management components, and signal/data handling units. A solar cell generates DC power which, when modulated by the laser's intensity, controls the motor's vibrations. A single laser source can drive multiple actuators alternately, with simultaneous control achieved by rapidly

sweeping the laser beam between solar cells. Though this reduces the maximum vibration intensity of each motor proportionally to the number of motors, it doesn't directly reduce haptic experience. Additionally, AC signals for an LRA motor are generated by scanning the laser over an additional solar cell connected oppositely, creating an AC signal without a PV inverter. Laser modulation adjusts the vibration amplitude, while laser sweeping speed sets the frequency of the signal, aligning it with the resonant frequency of the LRA for optimal performance. Transition times for signals generated this way are on par with those achieved by electronics, ensuring smooth, effective vibrations. Furthermore, simultaneous control of multiple LRAs can be achieved by rapidly scanning the laser across different solar cells, with the vibration intensity influenced by laser exposure duration and power.

Our prototype successfully leverages a laser source and a haptic device equipped with both ERM and LRA motors, connected to cost-effective PV cells. The PV cells, particularly efficient under high-power light sources, harness both visible light and part of the near-infrared spectrum. By employing two PV cells per direction in series, we could elevate the motors' output power, generating sufficient vibrations for various applications. The system utilizes a 445 nm blue laser diode with an output power of 500mW, carefully controlled to scan the PV cells. We ensured safety compliance by adhering to the Maximum Permissible Exposure limit, and balanced power density by increasing the beam size to match the PV cells' dimensions. Despite the varied detected vibration intensities and pattern shapes between the ERM and LRA due to distinct motor characteristics, our final implementation accommodates diverse vibration patterns. The potential applications of this system are extensive, including battery-free haptic gloves, ambient haptic feedback, and the adaptation of conventional input devices, opening the doors to more sustainable, battery-free, and versatile haptic systems.

3 CONCLUSION AND FUTURE WORK

This paper demonstrates the potential of a laser-based wireless power transfer and haptic control mechanism to drive vibrotactile devices, eliminating the need for traditional electronics and batteries. While we have showcased the feasibility of this approach, we also acknowledge that our current implementation has room for improvement in several areas that our future work will address. These include optimizing conversion efficiency, particularly through high-efficiency solar cells, and ensuring safe use with stronger lasers. We also plan to extend our method's compatibility to high voltage actuators like piezo actuators, and work on enabling precise tracking and user input detection. Our ongoing exploration aims not only to refine the versatility and applicability of this technique but also to advance sustainable technologies and reduce electronic waste, as our approach offers implications extending beyond vibrotactile devices to sustainable smart devices in both virtual and physical environments.

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